

Use Case Diagram

Study Buddy — Dual-Agent AI Education System

UCS503P · Software Engineering Project

STUDY BUDDY

Study Buddy System — Application Boundary



STUDENT

Onboarding & Teaching

Login / Sign Up

Upload Study Resources

Teach via Text

Teach via Voice

View Performance Dashboard



TEACHER AGENT

Knowledge & Assessment

Process Resources & Build Knowledge Base

Generate Exam Questions

Grade & Evaluate Answers



CHILD AGENT

Learning & Memory

Take Cold Baseline Test

Extract & Update Memory

Take Closed-Book Exam



Teacher Agent



Child Agent

«include»
«include»

«include»



Human actor — Student

Association (actor uses feature)



AI actor — Teacher Agent



«include» — always-triggered behaviour



AI actor — Child Agent



Coloured zone = features owned by that actor